

Youngjae Cho

Machine Learning Researcher

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youngjae-cho.github.io | GitHub | Google Scholar

My research focuses on **robust and data-efficient machine learning** under imperfect supervision, spanning preference optimization, active learning, Bayesian adaptation, and multimodal learning.

EDUCATION

KAIST | M.S., Industrial & Systems Engineering

Mar 2022 - Feb 2024

Advisor: Il-Chul Moon

KAIST | B.S., Industrial & Systems Engineering

Mar 2017 - Feb 2022

SELECTED PUBLICATIONS

NeurIPS 2026

Accepted

Learning Where It Matters: Geometric Anchoring for Robust Preference Alignment

Youngjae Cho, Jongsuk Kim, Ji-Hoon Kim

First author

Geometric anchoring for robust preference optimization; **+3.6 percentage points** in AlpacaEval 2.0 LC win rate over SimPO.

AAAI 2024

Make Prompts Adaptable: Bayesian Modeling for Vision-Language Prompt Learning with Data-Dependent Prior

Youngjae Cho, HeeSun Bae, Seungjae Shin, YeoDong Youn, Weonyoung Joo, Il-Chul Moon

First author

Bayesian prompt adaptation with data-dependent priors for vision-language learning with scarce data and distribution shift.

ICML 2023

Sharpness-Aware Active Learning

Yoon-Yeong Kim*, **Youngjae Cho***, JoonHo Jang, Byeonghu Na, Yeongmin Kim, Kyungwoo Song, Wanmo Kang, Il-Chul Moon

Co-first author

Sharpness-aware sample acquisition to improve generalization under a limited labeling budget.

* Equal contribution. Additional publications appear on page 2.

RESEARCH EXPERIENCE

Pyler

Research Scientist

Oct 2025 - Present

Preference optimization. Developed GAPO (NeurIPS 2026, accepted) and multimodal preference-optimization methods for Nemotron-Nano-12B-v2-VL using SimPO and Megatron-Bridge.

Auditable supervision. Recovered decision trees from 299K LLM/VLM reasoning traces; improved content-safety macro F1 from **0.777 to 0.857**.

Aiv Co.

ML Research Scientist

Mar 2024 - Oct 2025

Generative modeling. Led background-aware diffusion research for industrial anomaly detection. Evaluated on MVTec-AD and LOCO and deployed defect synthesis to improve production detector precision and recall.

KAIST

Graduate Research

Mar 2022 - Feb 2024

Data-efficient learning. Developed Bayesian vision-language prompt adaptation (APP, AAAI 2024; first author) and sharpness-aware active learning (SAAL, ICML 2023; co-first author).

ADDITIONAL PUBLICATIONS

arXiv 2024 Preprint	Background-Aware Defect Generation for Robust Industrial Anomaly Detection Youngjae Cho, Gwangyeol Kim, Sirojbek Safarov, Seongdeok Bang, Jaewoo Park First author
ICML 2022 Workshop	Improving Group-based Robustness and Calibration via Ordered Risk and Confidence Regularization Seungjae Shin, Byeonghu Na, HeeSun Bae, JoonHo Jang, Hyemi Kim, Kyungwoo Song, Youngjae Cho, Il-Chul Moon Workshop on Spurious Correlations, Invariance and Stability
IEEE SMC 2021	Predict Sequential Credit Card Delinquency with VaDE-Seq2Seq Yeongmin Kim, Youngjae Cho, Hanbit Lee, Il-Chul Moon

ACADEMIC SERVICE

ICML 2026 | Gold Reviewer

NeurIPS 2026 | Reviewer

AWARDS

NVIDIA Nemotron Hackathon

Track B Winner | 2026

Team winner, Domain-Specialized Model track.